



PAW Project



Backend (.NET) (30p)

-  **ASP.NET Core Web API (5p)**
 - required
 - Minimum 5 controllers – keep it RESTful!
 - Proper REST endpoints for smooth data flow.
-  **Entity Framework Core (10p)**
 - required
 - Minimum 6 entities to model your data.
 - **User** and **Role** are counted here
 - Relationships required:
 - One-to-Many (e.g., User has many Posts)
 - Many-to-Many (e.g., Posts and Tags)
 - Migrations must be used
-  **Authentication & Authorization (5p)**
 - ASP.NET Core Identity for secure logins.
 - Authorization with roles (minimum Admin + User) – protect those endpoints!
 - At least 2 endpoints restricted to Admin only.
-  **Exception Handling (3p)**
 - Global exception middleware or filter – handle errors gracefully.
-  **Logging (2p)**
 - Use built-in logging or Serilog – keep track of what's happening.
-  **Architecture (5p)**
 - DTOs for clean data transfer.
 - Service Layer for business logic.
 - Repository Pattern for data access.
 - Dependency Injection – keep things modular!



Frontend (Angular) (20p)

- Minimum 5 pages (5p)
 - required
 - login / register counts as pages
- Authentication (login/register) (4p) – make signup easy.
- Protected routes (2p) – secure your app.
- Forms with validation (2p)
- API integration (2p) – connect to your backend seamlessly.
 - required
- Folder structure (5p)

Note

Frontend can be Angular / React / Vue, or any of the cool frontend frameworks. Just make sure it is a web html frontend and don't let it be Vanilla!

Grading

- all the tasks totals to **50p**, but the maximum points one can score is **40p**, so each can choose what tasks to do for the maximum points (each can choose from the tasks that are not **required**)
- the **required** has to be present, otherwise those will be deducted
- project is 40% of the total grade, so the points system is created for **40p maximum**, to be easier to follow in the big overview.

Examples

1. 🤖 Event Management System

- **Entities**
 - User
 - Event
 - Ticket
 - Venue
 - Category
 - EventRegistration
- **Features**
 - Create events
 - Users register for events
 - Admin manages venues and categories
 - Event Capacity Limit

2. 📖 Online Course Platform

- **Entities**
 - Course
 - Lesson
 - Instructor
 - Student
 - Enrollment
 - Review
- **Features**
 - Students enroll in courses
 - Lessons inside courses
 - Reviews / Ratings
 - Admin manages courses

3. 🍔 Restaurant reservation system

- **Entities**
 - Restaurant
 - Table
 - Reservation
 - Menu / Menu Item
 - Category
 - Review

- **Features**

- Table booking
- Reservation management
- Menu browsing
- Reviews

4. 🏋️ **Fitness Tracker**

- **Entities**

- Workout
- WorkoutSession
- Exercise
- User
- Goal
- Progress

- **Features**

- User add workouts
- Exercise library
- Progress stats